

Analysis PhD Exam, May 1999

I

State the following theorems.

1. Hahn–Banach.
2. The Egorov theorem.
3. The Vitali convergence theorem.
4. The Radon–Nikodym theorem.
5. The Fubini theorem.
6. The monotone convergence theorem.
7. The completeness theorem for $L_F^p(\mu)$.
8. The equality $|g\mu| = |g| |\mu|$.

II

Prove one of the theorems 3 or 7.

III

Solve the following problems, where (X, Σ, μ) is a measure space and F is a Banach space.

1. If $f_n \rightarrow f$ in $L_F^p(\mu)$ and $f_n \rightarrow g$, μ -a.e., then $f = g$, μ -a.e.
2. Let $\mathcal{F} \subset \Sigma$ be a σ -algebra. Prove the existence of the conditional expectation $E(f \mid \mathcal{F})$ for $f \in L_F^p(\mu)$.
3. Let $\mathcal{R}$ be a ring generating Σ and $f \in L_F^1(\mu)$. Prove that if $\int_A f d\mu = 0$ for every $A \in \mathcal{R}$, then $f = 0$, μ -a.e.
4. Let $(X, \mathcal{F})$ and $(Y, \mathcal{T})$ be measurable spaces, and let $\mu : \mathcal{F} \rightarrow \mathbb{R}$ and $\nu : \mathcal{T} \rightarrow \mathbb{R}$ be σ -additive measures. Prove the existence of the product measure $\mu \times \nu : \mathcal{F} \times \mathcal{T} \rightarrow \mathbb{R}$ satisfying $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$ for $A \times B \in \mathcal{F} \times \mathcal{T}$, and $|\mu \times \nu| = |\mu| \times |\nu|$.
5. If $f \in L_F^1(\mu)$, there is a sequence (f_n) of $\mathcal{R}$ -step functions

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Note. Write complete computations and explanations. Do not use arrows or other unnecessary signs. Use words for explanations. Write complete sentences.